
Algorithm 2: PRUNE($p, \mathcal{E}, \alpha, R$)

Input: Graph G , point $p \in P$, candidate set $\mathcal{E} \subseteq P$, distance threshold $\alpha \geq 1$, degree bound R

Output: G modified by setting at most R new out-neighbors for p

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1  $\mathcal{E} \leftarrow (\mathcal{E} \cup N_{\text{out}}(p)) \setminus \{p\}$ 
2  $N_{\text{out}}(p) \leftarrow \emptyset$ 
3 while  $\mathcal{E} \neq \emptyset$  do
4    $p^* \leftarrow \arg \min_{p' \in \mathcal{E}} D(p, p')$ 
5    $N_{\text{out}}(p) \leftarrow N_{\text{out}}(p) \cup \{p^*\}$ 
6   if  $|N_{\text{out}}(p)| = R$  then
7     break
8   end
9   for  $p' \in \mathcal{E}$  do
10    if  $\alpha \cdot D(p^*, p') \leq D(p, p')$  then
11       $\mathcal{E} \leftarrow \mathcal{E} \setminus \{p'\}$ 
12    end
13  end
14 end
```
